American Computer Science League

2025 Finals ● Short Problems Solutions ● Junior Division

1. Boolean Algebra

$\overline{AB} + \overline{A}(B + \overline{A}) + \overline{\overline{A} + B} = \overline{A} + \overline{B} + \overline{A}\,B + \overline{A} + \overline{A}\,\overline{B}$
$= \overline{A}\,(1 + B + 1 + \overline{B}) + \overline{B}$
$= \overline{A} + \overline{B}$

B. $\overline{A} + \overline{B}$

2. Boolean Algebra

$\overline{A}\,\overline{B}\,C + \overline{A}\,B\,\overline{C} + A\,\overline{B}\,\overline{C} + \overline{A\,B\,C}$
$= \overline{A}\,\overline{B}\,C + \overline{A}\,B\,\overline{C} + A\,\overline{B}\,\overline{C} + \overline{A} + \overline{B} + \overline{C}$
$= \overline{A}(\overline{B}\,C + B\,\overline{C} + 1) + A\,\overline{B}\,\overline{C} + \overline{B} + \overline{C}$
$= \overline{A} + \overline{B}(A\,\overline{C} + 1) + \overline{C}$
$= \overline{A} + \overline{B} + \overline{C}$

To be FALSE: $\overline{A} = 0$, $\overline{B} = 0$ and $\overline{C} = 0$
So $A = 1, B = 1, C = 1$ or (1,1,1)
Therefore, 7 ordered triples make it TRUE.

Alternatively, the first 3 terms give solutions (0,0,1), (0,1,0), and (1,0,0). The last term yields everything but (1,1,1), which includes the other 3 solutions.

C. 7

3. Digital Electronics

$\overline{(\overline{\overline{AB} + B})B}$
$= \overline{\overline{AB}}\,\overline{B}\,B$ using DeMorgan's Law on the parenthesized expression
$= 0$ because $\overline{B}$ *and* B cancel out so $\overline{AB}$ can be ignored
$= 1$

D. 1

4. Digital Electronics

The Boolean expression for the digital circuit is: $\overline{A}\,\overline{((A + B)BC)}$

$\overline{A}\,\overline{((A + B)BC)}$
$= \overline{A}(\overline{A + B} + \overline{BC})$
$= \overline{A}(\overline{A}\,\overline{B} + \overline{B} + \overline{C})$
$= \overline{A}\,\overline{B} + \overline{A}\,\overline{B} + \overline{A}\,\overline{C}$
$= \overline{A}(\overline{B} + \overline{C})$
To be TRUE $\overline{A} = 1$, so $A = 0$ and $\overline{B} + \overline{C} = 1$
$\overline{B} + \overline{C} = 1 \Rightarrow B = 0$ and $C =$* which is $(0, 0, *)$ OR
$\overline{B} + \overline{C} = 1 \Rightarrow B =$* and $C = 0$ which is $(0, *, 0)$.
This gives you 3 that make it TRUE: (0,0,0), (0,0,1), (0,1,0).

B. 3

5. Bit-String Flicking

(LSHIFT-2 01101) OR (RCIRC-3 (10101 AND NOT 01101))
= (LSHIFT-2 01101) OR (RCIRC-3 (10101 AND 10010))
= (LSHIFT-2 01101) OR (RCIRC-3 (10000))
= (10100 OR 00010) = 10110

D. 10110

6. Bit-String Flicking

Let X = abcde
LHS = (LCIRC-2 X) AND (RSHIFT-1 00110) OR 01010
= (LCIRC-2 abcde) AND 00011 OR 01010
= cdeab AND 00011 OR 01010
= 000ab OR 01010
= 0101b
RHS = (RCIRC-3 11010)
= 01011
LHS = RHS → 0101b = 01011
So a = *, b = 1, c = *, d = *, e = *. The answer is *1***.

C. *1***

7. Recursive Functions

$f(17) = f(17 - 3) + 1 = f(14) + 1 = -3 + 1 = -2$
$f(14) = f(14/2) - 2 = f(7) - 2 = -1 - 2 = -3$
$f(7) = f(7 - 3) - 2 = f(4) + 1 = -2 + 1 = -1$
$f(4) = f(4/2) - 2 = f(2) - 2 = 0 - 2 = -2$
$f(2) = f(2/2) - 2 = f(1) - 2 = 2 - 2 = 0$
$f(1) = 2 * 1 = 2$
So $f(17) = -2$

B. -2

8. Recursive Functions

$f(12,5) = f(12-3,5+1) - 1 = f(9,6) - 1 = 6 - 1 = 5$
$f(9,6) = f(9-2,6-2) + 2 = f(7,4) + 2 = 4 + 2 = 6$
$f(7,4) = f(7-2,4-2) + 2 = f(5,2) + 2 = 2 + 2 = 4$
$f(5,2) = f(5-2,2-2) + 2 = f(3,0) + 2 = 0 + 2 = 2$
$f(3,0) = f(3-2,0-2) + 2 = f(1,-2) + 2 = -2 + 2 = 0$
$f(1,-2) = 1 * -2 = -2$

C. 5

9. Prefix/Infix/Postfix Notation

a b / a b + c / c 2 ^ − +
= (a b /) (a b +) c / (c 2 ^) − +
= (/ a b) ((+ a b) c /) (^ c 2) − +
= (/ a b) ((/ + a b c) (^ c 2) −) +
= (/ a b) (− / + a b c ^ c 2) +
= + / a b − / + a b c ^ c 2
= + / a b − / + a b c ^ c 2

D. + / a b − / + a b c ^ c 2

10. Prefix/Infix/Postfix Notation

− + # 2 3 * 4 + / 8 2 3 # 4 5
= − + (# 2 3) * 4 + (/ 8 2) 3 (# 4 5)
= − + 4 * 4 (+ 4 3) 18
= − + 4 (* 4 7) 18
= − (+ 4 28) 18
= − 32 18
= 14

B. 14

11. Computer Number System

$52425_8 = 101\ 010\ 100\ 010\ 101_2$
$= 101\ 0101\ 0001\ 0101_2$
$= 5\ \ 5\ \ 1\ \ 5_{16}$
$= 5515_{16}$

C. 5515

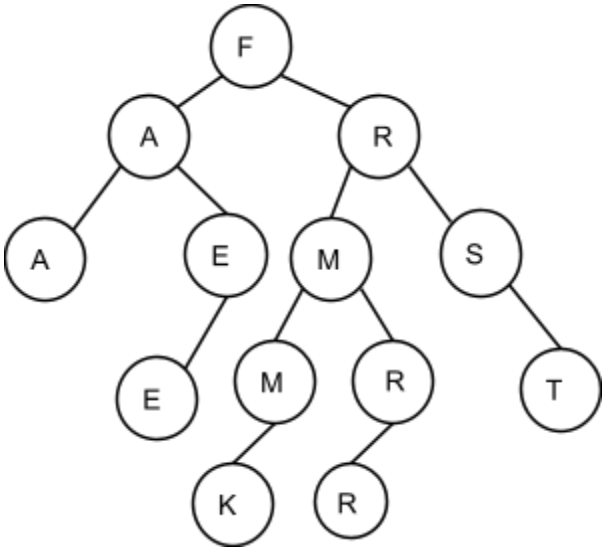
12. Computer Number Systems

Starting number: $2A_{16} = 42_{10}$
Last ticket number: $1A3_{16} = 419_{10}$
Number of tickets sold: 419 - 42 + 1 = 378
Cost of ticket: 1010_2 dollars = $10
Gate receipts: 378 x $10 = $3780
Alternatively, $1A3_{16} - 2A_{16} = 179_{16} + 1 = 17A_{16}$ so $17A_{16}$ = $1*256+7*16+10 = 378_{10} * 10_{10} = 3780_{10}$

C. 3780

13. Data Structures

The binary search tree for **FARMERSMARKET** is:



The nodes at depth 3 are E, M, R, T.

C. E M R T

14. Data Structures

The queue is constructed using FIFO as follows:

C CI I NIL L LA A AN ANT ANTR NTR NTRO TRO

C. T

15. Graph Theory

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}^2 = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 \\ 1 & 2 & 1 & 1 \end{bmatrix}$$

Therefore, there are 16 paths of length 4.

D. 16

16. Graph Theory

The cycles are: ABA, AFA, AGFA, AFCBA, AGFCBA, ABDGFA, BDGFCB, CFC. There are 8 of them.

C. 8

17. What Does This Program Do? (Branching)

In tracing the program, the last values of x, y, and z are 39, 16, 2

```
x = 21
y = 18
z = 2
if x > y + z then
   x = x + y
end if
if x - z > y then
   y = y - z
end if
output x - y * z
```

```
decision    x  y  z
            21
               18
                  2
21>20
            39
37>18
               16
39 - 16 * 2
```

Therefore, the output is 39 - 16 * 2 = 39 - 32 = 7.

A. 7

18. What Does This Program Do? (Loops)

```
for x = 1 to 50 step 3
   if x % 5 == 0 then
      output x
   end if
   if x % 4 == 0 then
      output x
   end if
   s = int(sqrt(x))
   if s*s == x then
      output x
   end if
next x
```

This program checks the values of x to find the multiples of 4 and 5 and the perfect squares.
x: 1, 4, 7, 10, 13, 16, 19, 22, 25, 28, 31, 34, 37, 40, 43, 46, 49
Divisible by 4: 4, 16, 28, 40
Divisible by 5: 10, 25, 40
Perfect squares: 1, 4, 16, 25, 49

The different numbers output are: 1, 4, 10, 16, 25, 28, 40, 49
There are 8 of them.

C. 8

19. What Does This Program Do? (Arrays)

```
for x = 0 to 8
   for y = x+1 to 9
      if arr(x) != 1 &&
         arr(y) % arr(x) == 0
         then
            arr(y) = 1
      end if
   next y
next x
```

This program performs Erastothenes' Sieve by eliminating the multiples of each number so only the primes numbers are left which are 2, 3, 5, 7, 11. There are 5 of them that are not 1.

B. 5

20. What Does This Program Do? (Strings)

```
ans = ""
s = "ACSL_IS_FUN."
for x = 0 to len(s)-1
   if s[x] >= "A" &&
         s[x] <=  "Z" then
      if s[x] < "M" then
         ch = shift(s[x],3)
      else
         ch = shift(s[x],-2)
      end if
      ans = ans + ch
   end if
next x
output ans
```

Each letter in the string is shifted 3 characters to the right if it is before "M" in the alphabet and 2 characters to the left otherwise. A becomes D; C becomes F; and S becomes Q. The final string once the letters have been concatenated from front to back is DFQOLQISL.

D. DFQOLQISL